

DENNIS FARMER

Ann Arbor, Michigan

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Education

University of Michigan - Ann Arbor, MI

Anticipated May 2027

B.S.E. Data Science - College of Engineering, Minor in Music

GPA: 3.87/4.00

Current Coursework: Web Systems, Regression Analysis, Music and AI, Digital Sound Synthesis

Coursework: Machine Learning, Computational Methods for Statistics, Data Structures and Algorithms,
Probability Theory, Linear Algebra, Discrete Mathematics

Projects

Shazam Music Recognition Clone - Project Lead

June 2025 – November 2025

Michigan Data Science Team

Ann Arbor, MI

- Replicated the Shazam algorithm in NumPy and integrated it into a React Native mobile app interface, with a web-scraped song library stored using SQLite, in collaboration with one other project co-lead
- Designed and taught an 8-week curriculum to teach project members the short-time Fourier transform, the Shazam algorithm, SQL, and React Native, for members to create their own version of Shazam

Sonification of Principal Component Analysis for Music Recommendation

December 2025

Digital Sound Synthesis - Final Project

Ann Arbor, MI

- Designed an interpretable content-based music recommender system, applying principal component analysis to MusiCNN feature embeddings of songs and selecting recommendations based on k-nearest neighbors of songs
- Organized results into an interactive dashboard, with a clickable PCA scatter plot to generate 5 second audio sonifications of songs for interpretability of the music recommender system (Plotly, Dash, ChuckK)
- Generated sonifications of songs by mapping principal component axes to synthesizer parameters, with parameterization guided by the correlation loadings between principal components and feature embeddings

Search Engine with MapReduce

November 2025

Web Systems - Final Project

Ann Arbor, MI

- Built a scalable full stack search engine using a service-oriented architecture
- Designed and implemented a multi-stage MapReduce pipeline executed with Hadoop for creating a segmented inverted index of 3064 crawled Wikipedia pages
- Created a REST API for distributed search query result retrieval, based on tf-idf text analysis and PageRank

Convolutional Neural Network for Image Classification

September 2023 – December 2023

Michigan Data Science Team

Ann Arbor, MI

- Designed a convolutional neural network architecture to distinguish between AI-generated faces and real human faces with a team of five people (PyTorch)
- Wrote a web application for uploading images and receiving classification results via API, with interpretability via GradCam convolutional layer visualizations (React, Express)

Research Experience

Machine Learning Researcher

May 2025 - Present

Cooperative Institute for Great Lakes Research

Ann Arbor, MI

- Utilizing PyTorch and HPC to recommend ideal buoy sensor placements to enhance weather forecasting
- Training convolutional gaussian neural processes on multi-modal environmental Great Lakes data for statistical downscaling and surface temperature uncertainty quantification (PyTorch)

Roaming Behavior of Domestic Cats

October 2024 - December 2024

Computational Methods for Statistics - Final Research Paper

Ann Arbor, MI

- Developed a beta regression model based on demographic variables to predict proportion of time domestic cats spent away from home, with negligible bias and MSE verified with Monte Carlo simulations
- Concluded cats roam farther away in dry conditions than during rainfall with a linear mixed-effects model, with bootstrapped 95% confidence interval for difference in means of (0.276 km, 1.069 km)

Skills

Languages: C++, Python (PyTorch), JavaScript (React), HTML, CSS, R, SQL, Bash, ChuckK

Tools and Software: Git, Docker, ngrok, Vim, Make, VS Code, GNU Debugger (gdb), Valgrind